

Chapter 3: Results

All results presented in this chapter are supported by reproducible code and computational outputs provided in the Appendix.

3.1 Principal Component Analysis

Principal Component Analysis (PCA) was conducted on the final processed dataset using Python to examine multivariate patterns in water chemistry across the defined distance groups (near, mid, and far). The analysis included 25 chemical parameters across 375 sampling observations. PCA was selected as a dimensionality reduction technique to summarize the variability in a high-dimensional dataset while preserving the underlying structure of relationships among variables (Reddy et al., 2020).

Prior to analysis, several preprocessing steps were applied to ensure consistency and comparability across variables. All chemical parameters were standardized using `StandardScaler()` to normalize differences in measurement units (e.g., mg/L, $\mu\text{g/L}$, and unitless pH), ensuring that no single variable disproportionately influenced the analysis. Due to the structure of the dataset, multiple rows corresponding to the same sampling event were aggregated based on site and sampling date, resulting in one observation per sampling event with a complete set of chemical variables. Missing values were addressed using median imputation, allowing all observations to be retained while reducing the influence of extreme or skewed values. The full implementation of PCA, including data aggregation by sampling event, median imputation, standardization, and extraction of principal components and loadings, is provided in Appendix C.

The proportion of variance explained by each principal component was used to assess how much information from the original dataset was captured. The first principal component (PC1) explained 21.4% of the total variance, while the second principal component (PC2) explained 14.63%. Together, these components accounted for 36.03% of the total variance as summarized in Table 3.1.1. This indicates that variability in water chemistry is distributed across multiple components rather than being concentrated along a single dominant gradient.

Table 3.1.1.

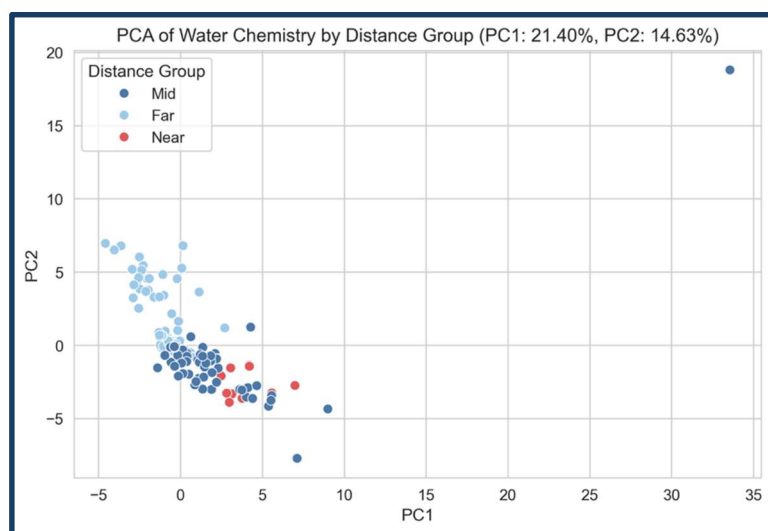
Proportion of variance explained by the first two principal components and their cumulative contribution to total dataset variability.

Component	Explained Variance (%)
PC1	21.4
PC2	14.63
Cumulative	36.03

Figure 3.1.1 presents the PCA scatter plot of PC1 and PC2, with sampling observations colored by distance group. The distribution of observations shows visible clustering patterns along the principal components. Samples from different distance groups occupy distinct regions of the plot, although there is noticeable overlap between groups, particularly between near- and mid-field observations. Far-field observations appear more dispersed across the PCA space, indicating a wider spread along both principal components.

Figure 3.1.1.

Principal Component Analysis (PCA) scatter plot showing sampling observations projected onto the first two principal components (PC1 and PC2). Points are colored by distance group (Near, Mid, Far).



To further examine the contribution of individual variables to each component, the loadings for PC1 and PC2 were calculated. Loadings represent the strength and direction of each variable's contribution to a principal component. Selected variables with the highest absolute loadings are presented in Table 3.1.2, and their relative contributions are illustrated in Figure 3.1.2.

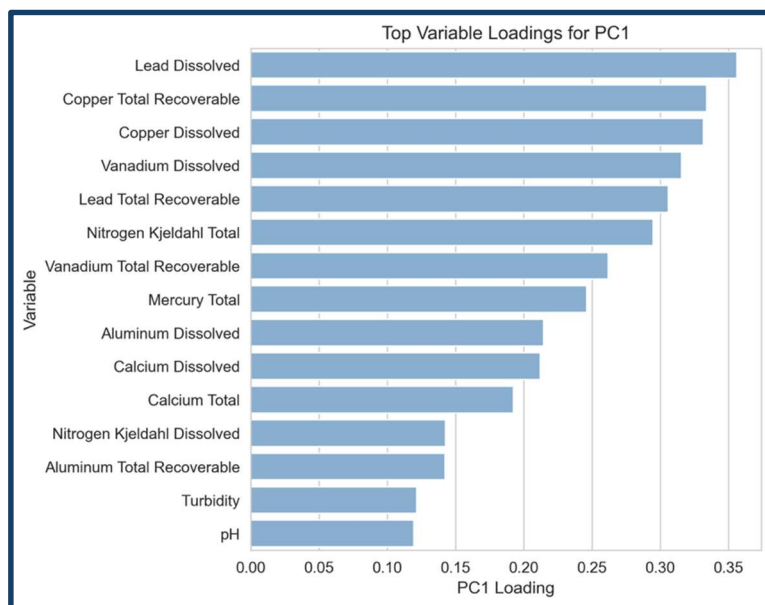
Table 3.1.2.

Principal component loadings for selected variables with the highest absolute contributions to PC1 and PC2.

Variable	PC1	PC2
Nickel Dissolved	-0.0434680986381497	0.38583397545001000
Nickel Total Recoverable	0.025844969909527400	0.37519438638754000
Lead Dissolved	0.3561335762980570	0.2555408122928200
Copper Total Recoverable	0.3340929899989810	0.29991184395881400
Copper Dissolved	0.3317280335775710	0.29604653580706900
Vanadium Dissolved	0.315689034559603	-0.19101690342155800
Lead Total Recoverable	0.30606343322521100	0.2595246968208510
Nitrogen Kjeldahl Total	0.2948195275870230	-0.13725897421734300
Vanadium Total Recoverable	0.2619834944984720	-0.1811424318429190
Mercury Total	0.24612609513575900	0.057279639814746300
Nitrogen Kjeldahl Dissolved	0.14258773213469500	-0.24185666471553800
Aluminum Dissolved	0.21471170345914400	-0.006865394778270850
Calcium Dissolved	0.21216393787034900	-0.21458172002301900
Calcium Total	0.19252659887784400	-0.2052896761653070
pH	0.11959899697471800	-0.1869226003483150
Turbidity	0.12171671599524500	-0.18687816847953200
Methyl Mercury	0.10178959914587400	-0.15712644466494900
Oxygen Dissolved (Field Meter)	-0.09735767875693580	0.15578895700472300
OXYGEN BIOCHEMICAL DEMAND	0.09302600486670800	-0.14305813671314900
Aluminum Total Recoverable	0.1423210358303610	-0.053023762547368400
Oxygen dissolved % saturation	0.08336021618885830	-0.10614170559770700
Mercury Dissolved	0.0719586406334068	-0.039128761836278400
Phosphorus Total Dissolved	0.0308097384209908	0.059760294242138200
Phosphorus Total	-0.005217430791922860	-0.05913135463416890

Figure 3.1.2.

Variable loadings for the first principal component (PC1), showing the relative contribution of water chemistry variables to the dominant gradient of variation.



Variables contributing most strongly to PC1 include lead dissolved (0.356), Copper total recoverable and dissolved (0.334,0.331), and vanadium dissolved (0.316), all of which exhibit positive relationships with the component. In contrast, nickel dissolved (-0.236) contributes negatively to PC1. For PC2, the strongest contributors include nickel dissolved (0.385), nickel total recoverable (0.375) and lead dissolved (0.255). Negative loadings are also observed, such as calcium dissolved (-0.215), indicating an inverse contribution along the second principal component. The presence of both positive and negative loadings reflects the multidirectional structure of variability captured by the PCA.